# Data Science 500 Assignment

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Name and Surname: Thego Malema

Student ITS No: 402311912

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Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

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**Declaration** _ _ _

## Declaration

I Thego Pheaha Malema, student number 402311912 hereby declare that this project report titled **"Data Science 500 Project Report: Web Scraping, Exploratory Data Analysis, and Predictive Modeling on Books to Scrape"** is my own original work.

- I confirm that this material has not been submitted previously for any other academic module, qualification, or institution.
- All secondary sources, libraries, and online tools utilized have been properly acknowledged and referenced in accordance with the Harvard Referencing style guidelines.
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## Introduction

In the modern digital era, the proliferation of online data has transformed how organizations and researchers extract actionable insights. Web scraping has emerged as a fundamental data acquisition technique, enabling the automated collection of unstructured data from web pages for downstream analytical and predictive tasks. This project report details an end-to-end data science pipeline executed for the Data Science 500 module, centered around the systematic acquisition, cleaning, exploration, and modeling of online product data.

The primary objective of this project is to harvest product information from the sandbox e-commerce website, *Books to Scrape*, transform the raw scraped text into a structured analytical dataset using Python, and perform exploratory data analysis (EDA) to uncover underlying market trends. Furthermore, the project investigates the feasibility of predictive analytics by implementing a Linear Regression model to evaluate whether product features—specifically star ratings and stock availability—can reliably predict book prices.

This report is structured into several key phases:

- **Data Acquisition:** Utilizing Python libraries such as `requests` and `BeautifulSoup` to scrape multi-page product listings.
- **Data Cleaning & Preparation:** Employing the `pandas` library to handle missing values, convert data types, and normalize currency and rating formats into a clean CSV dataset.
- **Exploratory Data Analysis & Visualization:** Utilizing `matplotlib` and `seaborn` to visualize price distributions, rating frequencies, and feature correlations.
- **Predictive Modeling:** Training and evaluating a Linear Regression model to test relationships between variables, followed by a discussion of model performance and limitations.

## Data Acquisition & Web Scraping

### *Overview of the Data Source*

To build a functional and reproducible machine learning pipeline, data was harvested from **Books to Scrape** (*http://books.toscrape.com*), a simulated e-commerce web page specifically designed for web scraping practice. This platform provides static HTML structures containing product details, making it an ideal environment for extracting structured tabular data using Python.

### *Tools and Libraries*

The data acquisition pipeline was developed in Python within a Jupyter Notebook environment using the following core libraries:

- `requests`: Utilized to send HTTP `GET` requests to target URLs and retrieve raw HTML response content from the web server.
- `BeautifulSoup` **(from** `bs4`**)**: Employed to parse the raw HTML text, navigate the Document Object Model (DOM) tree, and locate specific target elements using CSS selectors.
- `pandas`: Used temporarily during extraction to compile individual records into structured DataFrames before export.

### *Scraping Methodology & HTML Parsing*

The target web page structure organizes each product under an article container with the CSS class `product_pod`. Within each container, the required attributes were targeted using specific HTML tags and classes:

1. **Book Title:** Extracted from the `title` attribute inside the anchor tag nested within the heading element (`h3 a`).
2. **Price:** Located within the paragraph element with the class `p.price_color`, which required string slicing to remove currency symbols (e.g., "£").
3. **Star Rating:** Extracted via the class attribute of the paragraph element (`p.star-rating`), where ratings are represented as string text (e.g., "Three", "Four") requiring mapping into numerical integer values.
4. **Stock Availability:** Extracted from the availability class (`p instock availability`), capturing stock status attributes.

***Pagination and Scale***

To ensure a sufficient sample size for exploratory analysis and modeling, the scraping script was engineered with a dynamic loop to iterate across multiple catalog pages. By targeting the pagination elements, the script systematically crawled **10 pages**, successfully extracting a robust dataset comprising **200 individual book records**. The raw data was compiled and prepared for the subsequent data cleaning and transformation phase.

## Data Cleaning & Preparation

### *Initial Data Inspection*

Following the data acquisition phase, the raw scraped records were loaded into a `pandas` DataFrame for initial inspection. An exploratory examination of the raw data revealed several structural inconsistencies typical of web-scraped text:

- Price fields contained embedded currency symbols (£) and string data types, preventing numerical calculations.
- Star ratings were stored as categorical text strings (e.g., "Three", "Four") instead of integers.
- Columns required validation to check for missing values, null entries, or duplicate records that could skew downstream exploratory analysis and modeling.

### *Data Transformation and Cleaning Steps*

To prepare the dataset for numerical analysis and machine learning, the following transformation steps were executed programmatically using Pandas:

1. **Price Cleansing (`Price_GBP`):** The currency symbol was stripped from each string entry, and the resulting string values were cast into floating-point numbers (`float64`) to enable statistical computations.
2. **Rating Encoding (`Star_Rating`):** A custom mapping dictionary was implemented to convert textual star ratings into discrete numerical integers (i.e., *"One"* $\rightarrow$ $1$, *"Two"* $\rightarrow$ $2$, *"Three"* $\rightarrow$ $3$, *"Four"* $\rightarrow$ $4$, *"Five"* $\rightarrow$ $5$). This numeric encoding allows the feature to be utilized effectively within correlation matrices and regression models.
3. **Handling Missing and Duplicate Values:** The DataFrame was checked for missing values across all columns. No significant null values were detected; however, duplicate entries based on book titles were filtered out to ensure data integrity and uniqueness across all 200 records.
4. **Stock Availability Formatting:** The availability column was cleaned to remove extraneous whitespace and newline characters, standardizing the text strings.

### *Final Dataset Export*

Once all data cleaning and transformation procedures were successfully completed, the refined DataFrame was exported to a structured comma-separated values file named

books_to_scrape_clean.csv. This clean dataset served as the foundational input for the exploratory data analysis and predictive modeling phases.

## Data Visualization & Exploratory Data Analysis

### *Overview of Exploratory Data Analysis (EDA)*

To understand the underlying patterns, distributions, and relationships within the scraped book dataset, Exploratory Data Analysis (EDA) was conducted using **Matplotlib** and **Seaborn**. Visualizing the data is a critical step in data science pipelines

to identify trends, skewness, and potential anomalies before deploying predictive models.

***Key Visualizations Generated***

Three core visualizations were generated and analyzed within the Jupyter Notebook workflow:

**Distribution of Book Prices (**`histplot` **with KDE):**

- *Visualization Details:* A histogram paired with a Kernel Density Estimate (KDE) curve plotting the frequency of book prices in British Pounds (£).
- *Observation:* The price distribution exhibited a right-skewed pattern, indicating that the majority of books are clustered within lower price ranges, with fewer high-priced titles extending the tail.

**Price Distribution Across Star Ratings (**`boxplot`**):**

- *Visualization Details:* Box plots illustrating the spread, median, and interquartile ranges of book prices segmented by their star ratings (1 through 5 stars).
- *Observation:* The median prices across different star ratings remained relatively stable, suggesting that a book's star rating does not inherently command a higher or lower retail price on the platform.

**Frequency of Books per Star Rating (**`countplot`**):**

- *Visualization Details:* A categorical count plot showing the absolute frequency of books belonging to each star rating category.
- *Observation:* The counts across the rating tiers provided insight into how inventory is distributed across quality tiers in the sample.

***Summary of EDA Findings***

The exploratory data analysis revealed that while price variations exist across the dataset, simple categorical features such as star ratings do not show strong visual linear separation or correlation with pricing. This justified moving forward to quantitative predictive modeling to rigorously test these feature relationships mathematically.

## Predictive Modeling & Evaluation

### *Model Selection & Feature Engineering*

To determine whether quantitative relationships exist between book features and their retail prices, a supervised machine learning approach was implemented. **Linear Regression** was selected as the predictive model to evaluate if a linear function can accurately estimate book prices (`Price_GBP`) based on independent predictor variables.

Based on the available features in the cleaned dataset (books_to_scrape_clean.csv), the feature matrix ($X$) was constructed using:

- Star_Rating (numerical integer encoding from 1 to 5)
- 
- Stock availability / quantity attributes
- 

The target variable ($y$) was defined as the book's price in British Pounds (Price_GBP).

***Training and Testing Split***

To prevent overfitting and ensure an unbiased evaluation of the model's predictive capability on unseen data, the dataset was partitioned into training and testing subsets using an **80/20 split**:

- **Training Set:** 80% of the data was used to fit the Linear Regression coefficients.
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- **Testing Set:** 20% of the data was held out exclusively to evaluate model performance and generalizability.
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The scikit-learn (sklearn.linear_model.LinearRegression) library was utilized to instantiate, train, and test the model pipeline within the Jupyter Notebook environment.

***Model Evaluation Metrics***

To quantify the performance of the trained Linear Regression model on the test set, standard regression evaluation metrics were computed:

1. **Mean Squared Error (MSE):** Measures the average squared difference between the estimated values and the actual value.

2. **Root Mean Squared Error (RMSE):** Represents the square root of the variance of the residuals, providing error metrics in the original unit of measurement (GBP).

3. **R-Squared ($R^2$) Score:** Represents the proportion of variance in the dependent variable that is predictable from the independent variables.

***Results and Interpretation***

Upon evaluating the model on the test set, the Linear Regression model yielded an **$R^2$ score of $-0.0085$**.

- **Interpretation of Findings:** An $R^2$ score at or below zero indicates that the Linear Regression model performs worse than a horizontal baseline model (i.e., predicting the mean price for every book). This mathematical outcome demonstrates that, within this specific sandbox dataset, a book's star rating and stock status have little to no linear correlation with its retail price.

- **Critical Evaluation & Limitations:** This result highlights an important data science takeaway: not all features possess predictive power. On *Books to Scrape*, pricing is likely determined by uncaptured categorical variables (such as book genre, category metadata, or page count) rather than customer ratings alone. This insight provides valuable context for future model iterations, suggesting that text vectorization of descriptions or multi-feature categorization would be required to improve predictive accuracy.

## Conclusion & Recommendations

### *Project Conclusion*

This project successfully executed an end-to-end data science pipeline for the Data Science 500 module, taking raw web data from *Books to Scrape* through automated extraction, structured cleaning, exploratory analysis, and predictive modeling.

- Through **BeautifulSoup**, 200 book records were successfully harvested.

- Using **Pandas**, the data was cleaned, null values and duplicates were handled, prices were cast to floats, and star ratings were numerically encoded.

- **Matplotlib** and **Seaborn** visualizations revealed insights into price distributions and feature relationships.

- Finally, the implementation of a **Linear Regression model** yielded an $R^2$ score of $-0.0085$, proving mathematically that star ratings and stock indicators alone do not linearly predict book prices on this platform.

***Recommendations for Future Work***

To build upon this project and improve predictive performance in future iterations, the following steps are recommended:

1. **Feature Expansion:** Scrape additional descriptive attributes from individual product pages, such as book category/genre, number of reviews, or product description length, which may have stronger correlations with pricing.

2. **Advanced Modeling Techniques:** Implement non-linear machine learning models (such as Decision Trees, Random Forests, or Gradient Boosting) that can capture complex, non-linear relationships between variables.

3. **Scale Data Collection:** Expand the scraping script to capture the entire catalog across all available pages rather than a 10-page sample, providing a larger dataset for model training and cross-validation.

## Bibliography

Books to Scrape (2026) *Books to Scrape: Sandbox Web Scraping Site*. Available at: http://books.toscrape.com (Accessed: 22 October 2026).

Hunter, J.D. (2007) 'Matplotlib: A 2D graphics environment', *Computing in Science & Engineering*, 9(3), pp. 90–95. Available at: https://matplotlib.org (Accessed: 22 October 2026).

McKinney, W. (2018) *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython*. 2nd edn. Sebastopol, CA: O'Reilly Media. Available at: https://pandas.pydata.org (Accessed: 22 October 2026).

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V. and Vanderplas, J. (2011) 'Scikit-learn: Machine Learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830. Available at: https://scikit-learn.org (Accessed: 22 October 2026).

Reitz, K. (2026) *Requests: HTTP for Humans (v2.34.2)*. Read the Docs. Available at: https://requests.readthedocs.io (Accessed: 22 October 2026).

Richardson, L. (2025) *Beautiful Soup Documentation (v4.x)*. Crummy. Available at: https://www.crummy.com/software/BeautifulSoup/bs4/doc/ (Accessed: 22 October 2026).

Waskom, M.L. (2021) 'Seaborn: statistical data visualization', *Journal of Open Source Software*, 6(60), p. 3021. Available at: https://seaborn.pydata.org (Accessed: 22 October 2026).